RAJALAKSHMI ENGINEERING COLLEGE

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R2

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|---|---|
| **Started on** | Thursday, 6 August 2026, 10:19 AM |
| **State** | Finished |
| **Completed on** | Thursday, 20 August 2026, 11:40 AM |
| **Time taken** | 14 days 1 hour |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include <stdio.h>
int main(){
    int n,i;
    int count = 0;
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        count += 2;
        if(n%i==0)
        {
            count ++;

            }
        }


      count++;
        printf("%d",count);
        return 0;

}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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